

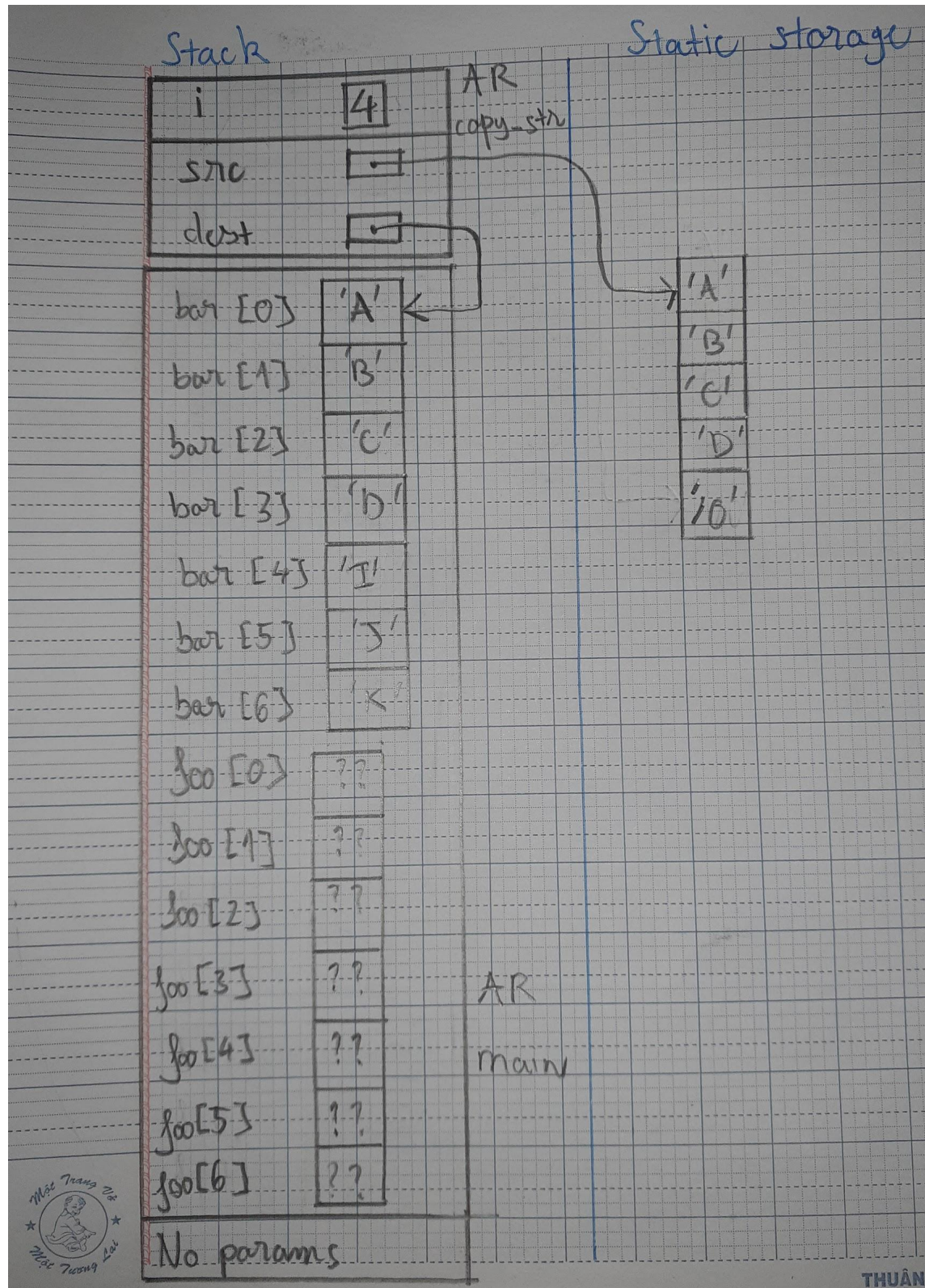
Duc Nguyen - UCID: 30102065

Lab section B01

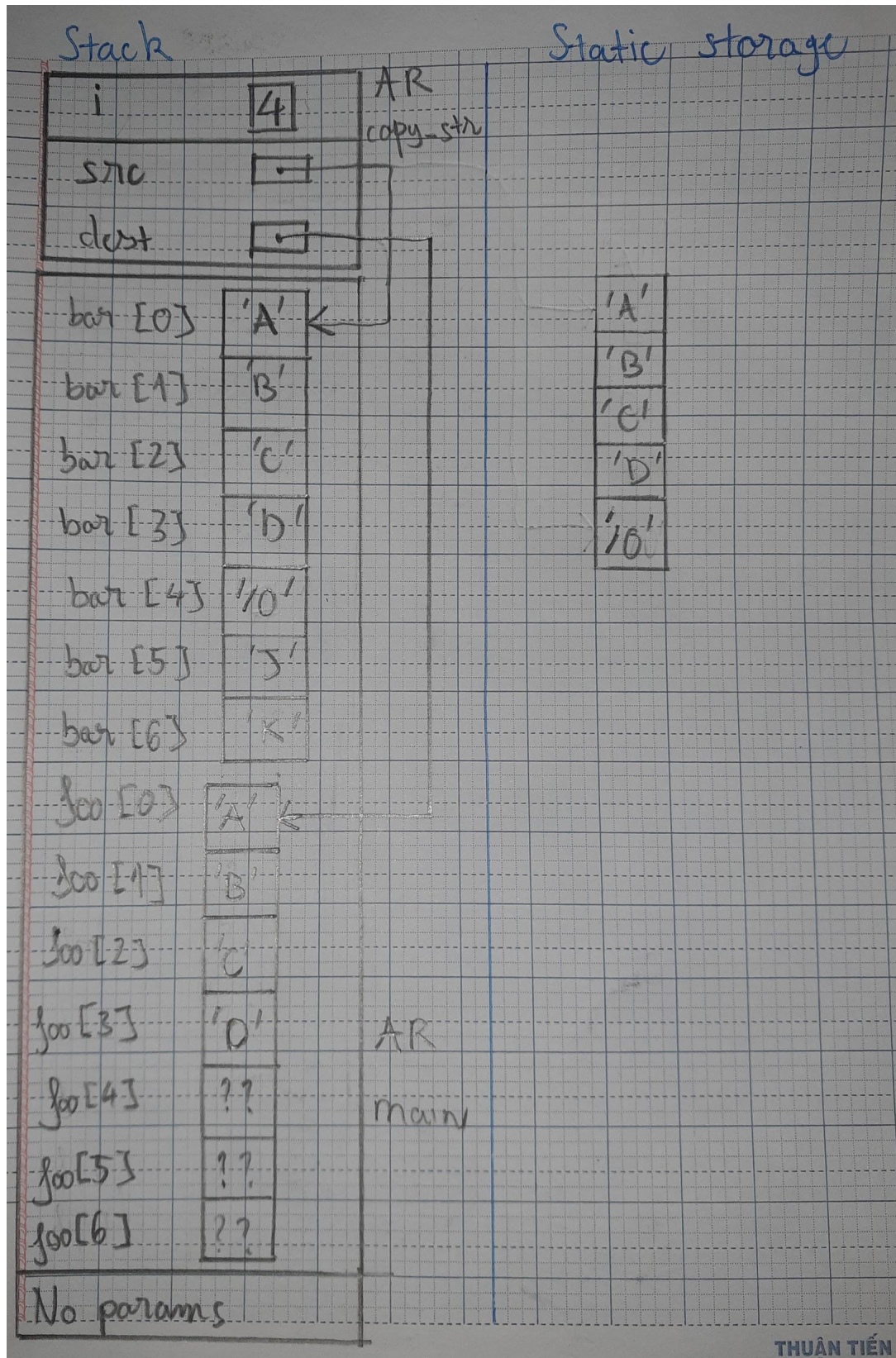
ENCM 335 - LAB 4

Exercise A:

1st time the program reaches point 1:

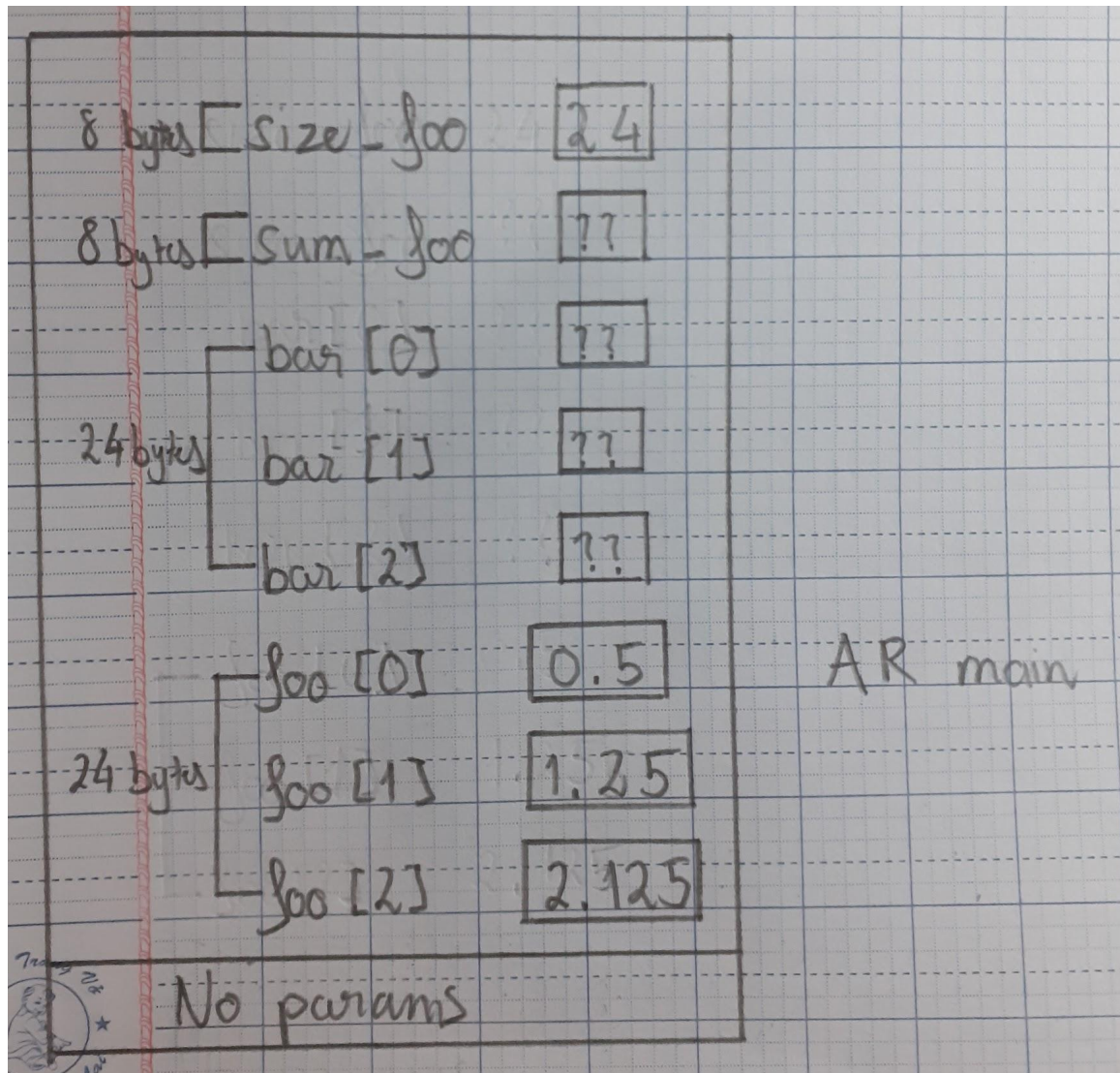


2nd time the program reaches point 1:

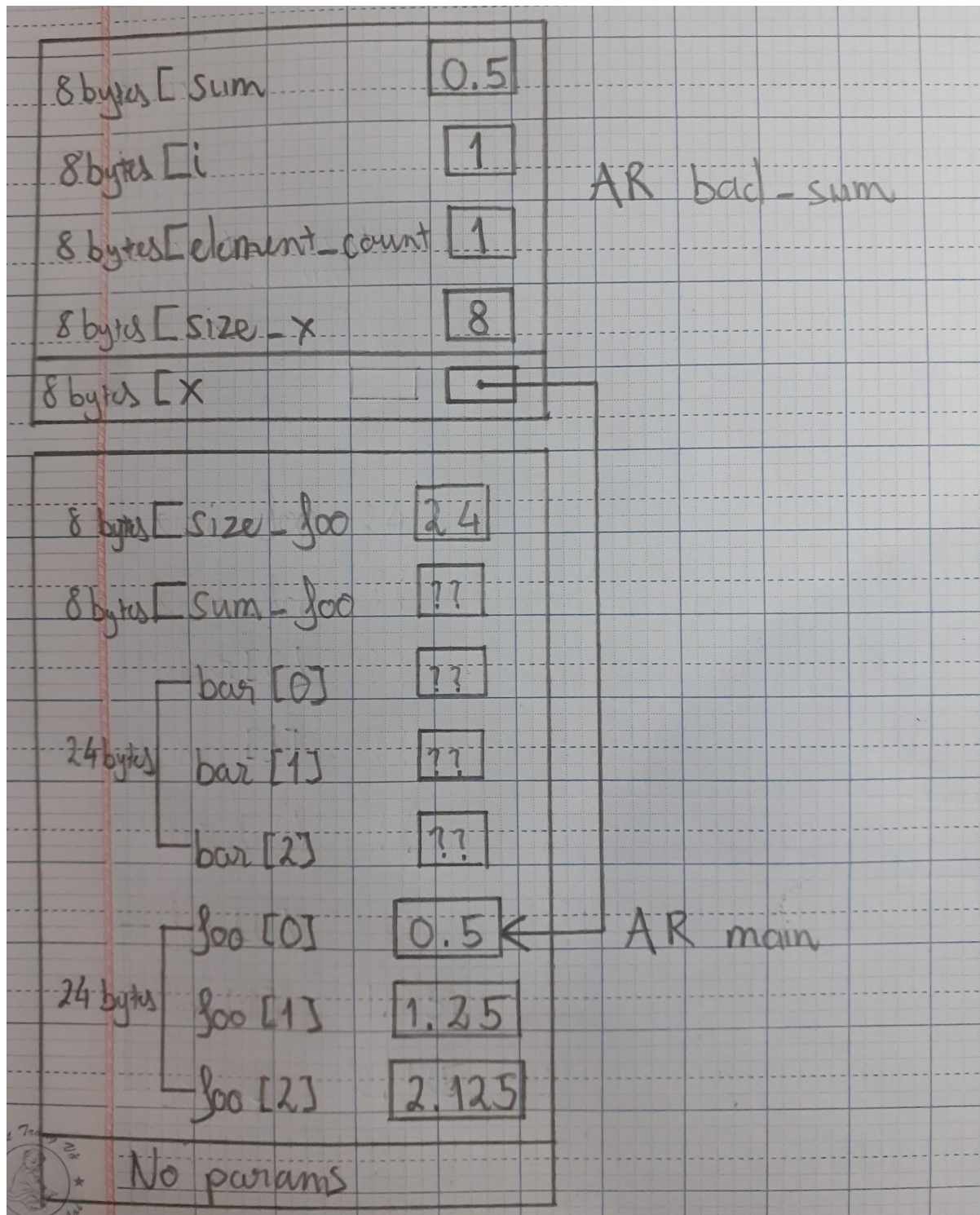


Exercise B:

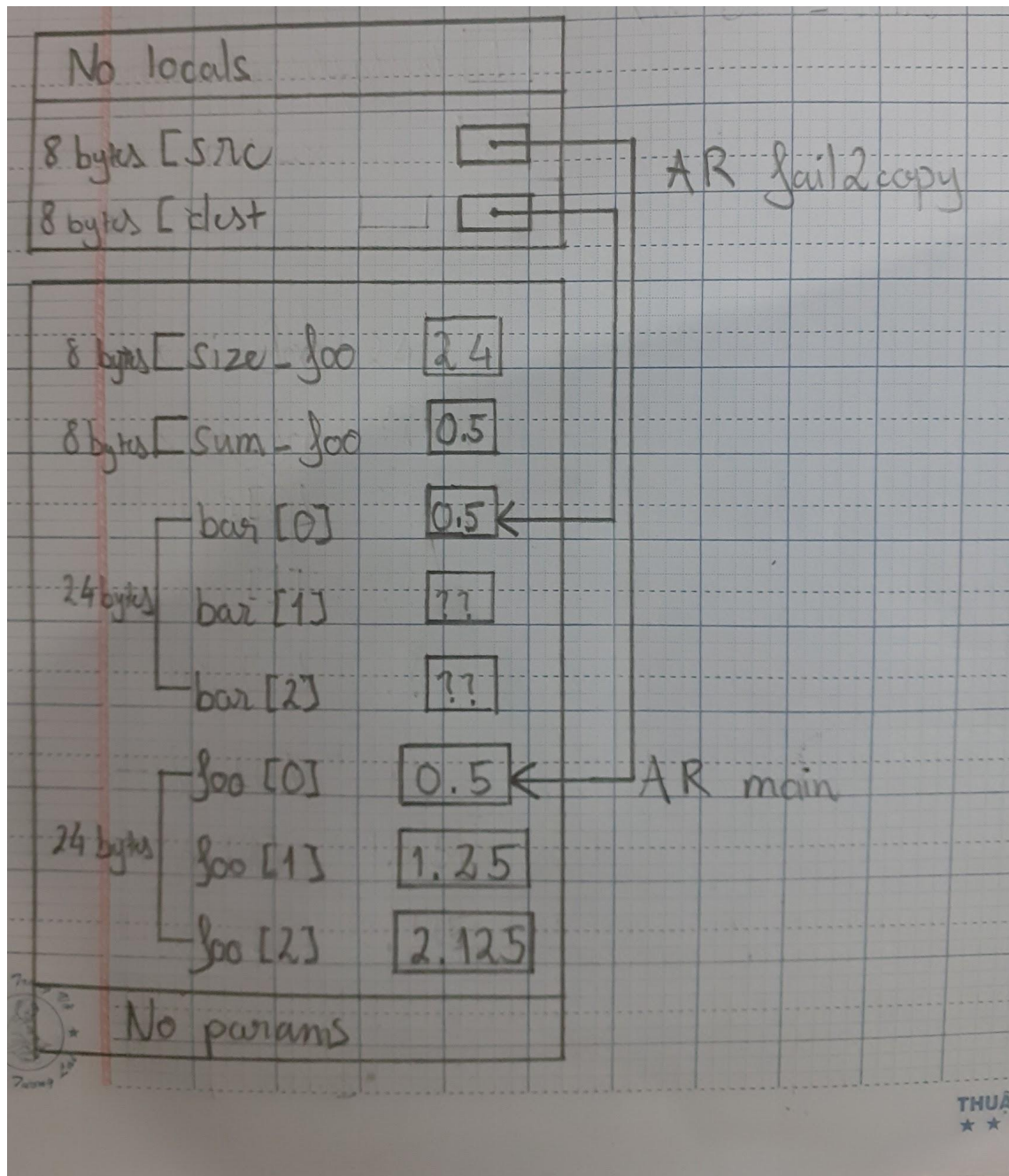
AR at point 1:



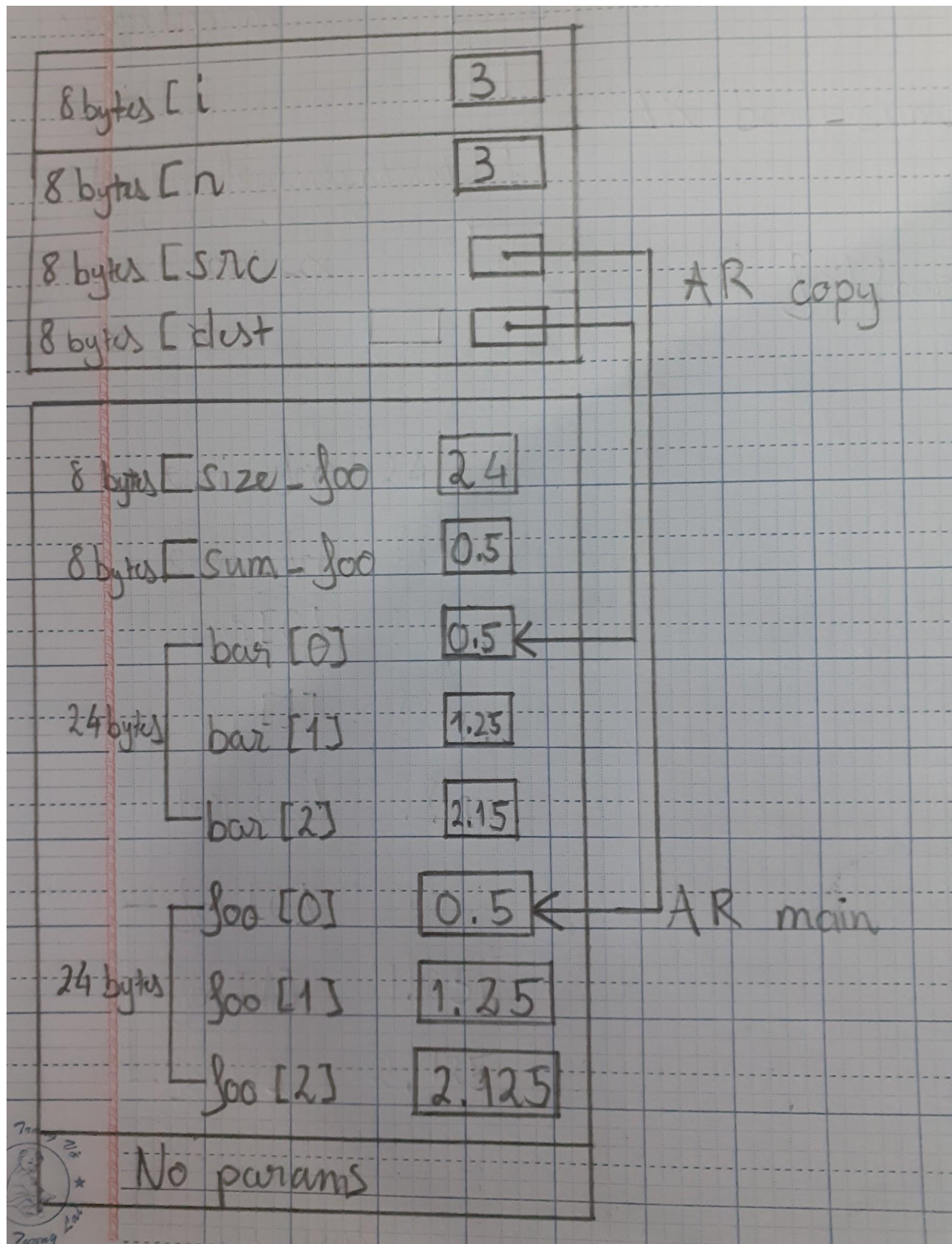
AR at point 2:



AR at point 3:



AR at point 4:



Exercise C:

Code:

```
1 // ENCM 335 Fall 2021 Lab 4 Exercise C
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char buffer[80]; // enough space for a string of length <= 79
8
9     // THIS IS A GOOD WAY TO LEARN SOMETHING ABOUT C STRINGS, BUT IT'S
10    // NOT A GOOD EXAMPLE OF READABLE OR PRACTICAL CODE!
11
12    // Put characters into the string using ASCII codes.
13    buffer[0] = 73;
14    buffer[1] = 110;
15    buffer[2] = 32;
16    buffer[3] = 67;
17    buffer[4] = 32;
18    buffer[5] = 116;
19    buffer[6] = 104;
20    buffer[7] = 101;
21    buffer[8] = 32;
22    buffer[9] = 118;
23    buffer[10] = 97;
24    buffer[11] = 108;
25    buffer[12] = 117;
26    buffer[13] = 101;
27    buffer[14] = 32;
28    buffer[15] = 111;
29    buffer[16] = 102;
30    buffer[17] = 32;
31
32    buffer[18] = 57;
33    buffer[19] = 32;
34    buffer[20] = 42;
35    buffer[21] = 32;
36    buffer[22] = 55;
37    buffer[23] = 32;
38    buffer[24] = 37;
39    buffer[25] = 32;
40    buffer[26] = 49;
41    buffer[27] = 49;
42    buffer[28] = 32;
43    buffer[29] = 105;
44    buffer[30] = 115;
45    buffer[31] = 32;
46    buffer[32] = 56;
47    buffer[33] = 46;
48    // Put the end-of-string character at the end of the string.
49    buffer[34] = 0;
50
51    printf("The string in buffer is \"%s\\n\"", buffer);
52    return 0;
53 }
```

Output:

```
monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab4
$ gcc -Wall lab4exC.c

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab4
$ ./a.exe
The string in buffer is "In C the value of 9 * 7 % 11 is 8."

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab4
$ |
```

Exercise D:

Code:

```
1 // ENCM 335 Fall 2021 Lab 4 Exercise D
2
3 #include <stdio.h>
4 #include <string.h>
5
6 int safecat(char *dest, const char* src, int dest_size);
7 // REQUIRES
8 //   dest_size > 0.
9 //   Array elements dest[0] ... dest[dest_size-1] exist.
10 //   dest points to the beginning of a C string with
11 //   length < dest_size.
12 //   src points to the beginning of a C string.
13 // PROMISES
14 //   If the sum of the lengths of the two strings is less than
15 //   dest_size, the string in the dest array is the concatenation
16 //   of the original string in dest and the string from src, and
17 //   the return value is the length of the new string.
18 //
19 //   If not, the string in dest is the concatenation of the original
20 //   string in dest and as many characters from src as possible,
21 //   while leaving room for '\0' in dest[dest_size-1], and the return
22 //   value is -1.
23
24 int main(void)
25 {
26     int rv;
27     char buf[10];
28
29     buf[0] = '\0';
30     rv = safecat(buf, "", 10);
31
32     printf("buf contains \"%s\" and rv is %d (expect \"\" and 0).\n",
33           buf, rv);
34     rv = safecat(buf, "abcde", 10);
35     printf("buf contains \"%s\" and rv is %d (expect \"abcde\" and 5).\n",
36           buf, rv);
37     rv = safecat(buf, "fghi", 10);
38     printf("buf contains \"%s\" and rv is %d (expect \"abcdefghi\" and 9).\n",
39           buf, rv);
40
41     buf[0] = '\0';
42     rv = safecat(buf, "jkl", 10);
43     printf("buf contains \"%s\" and rv is %d (expect \"jkl\" and 3).\n",
44           buf, rv);
45     rv = safecat(buf, "mnopqrs", 10);
46     printf("buf contains \"%s\" and rv is %d (expect \"jklmnopqr\" and -1).\n",
47           buf, rv);
48
49     buf[0] = '\0';
50     rv = safecat(buf, "01", 10);
51     printf("buf contains \"%s\" and rv is %d (expect \"01\" and 2).\n",
52           buf, rv);
53     rv = safecat(buf, "2345", 10);
54     printf("buf contains \"%s\" and rv is %d (expect \"012345\" and 6).\n",
55           buf, rv);
56     rv = safecat(buf, "6789ABCDEF", 10);
57     printf("buf contains \"%s\" and rv is %d (expect \"012345678\" and -1).\n",
58           buf, rv);
59     return 0;
60 }
61
62 int safecat(char *dest, const char* src, int dest_size)
63 {
64     int i, j;
65
66     if (strlen(dest)+strlen(src)<dest_size) {
67         for(i=strlen(dest), j=0; j<strlen(src); i++, j++)
68             dest[i]=src[j];
69         dest[i]='\0';
70         return strlen(dest);
71     }
72     else{
73         for(i=strlen(dest), j=0; i<dest_size-1; i++, j++)
74             dest[i]=src[j];
75         dest[i]='\0';
76         return -1;
77     }
78 }
```

Output:

```
monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab4
$ ./a.exe
buf contains "" and rv is 0 (expect "" and 0).
buf contains "abcde" and rv is 5 (expect "abcde" and 5).
buf contains "abcdefghi" and rv is 9 (expect "abcdefghi" and 9).
buf contains "jkl" and rv is 3 (expect "jkl" and 3).
buf contains "jklmnopqr" and rv is -1 (expect "jklmnopqr" and -1).
buf contains "01" and rv is 2 (expect "01" and 2).
buf contains "012345" and rv is 6 (expect "012345" and 6).
buf contains "012345678" and rv is -1 (expect "012345678" and -1).
```